

Code-Layout, Readability and Reusability

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used consistent Python indentation throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project files are organized into separate folders for models, datasets, and application code.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables and functions use descriptive names for better understanding.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are clearly named based on their purpose.
5	Code Comments	Inline and block comments explain logic	partial	Comments explain important logic and functionality.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable functions/modules for emotion detection and recommendation.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed duplicate and unnecessary code to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented to prevent application crashes.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Emotion Detection Module	Python	Emotion analysis workflow	High
2	AI Recommendation Module	Python	Personalized learning suggestions	High
3	Streamlit UI Components	Streamlit	User interface pages	Medium
4	Subject Recommendation Logic	Python	Multiple subjects	High
5	Utility Functions	Python	Input processing & validation	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Clear naming conventions and easy-to-understand logic.
Reusability	5	Core modules can be reused in similar AI-based education projects.
Documentation / Comments	4	Sufficient comments provided for important sections.
Overall Score	4	Clean, modular, and maintainable codebase.